



Food Cold Chain: Challenges and Opportunities for Digitalization

Abstract:

The food supply chain from 'Farm to Fork' involves multiple entities as the food items get harvested, processed, transported, and distributed. Food items that require refrigerated transport and storage get spoiled in the supply chain or are contaminated when ideal temperature and humidity conditions are not maintained. This white paper examines the key challenges involved in a cold supply chain and opportunities for remote monitoring to reduce food spoilage and contamination.

Authors:

Vivek Madathil (Chief Architect, Digital Business),
Shweta Naik (Project Manager, Digital Business),
Sasken Technologies Ltd.



Table of Content

Problem Statement	4
Typical Refrigerated Food Supply Chain	5
Challenges in the Refrigerated Food Supply Chain	7
FSMA 204	8
Understanding of GS1 Standard	9
Opportunities for Digitalization	10
References:	11
About the Authors	11
About Sasken	11



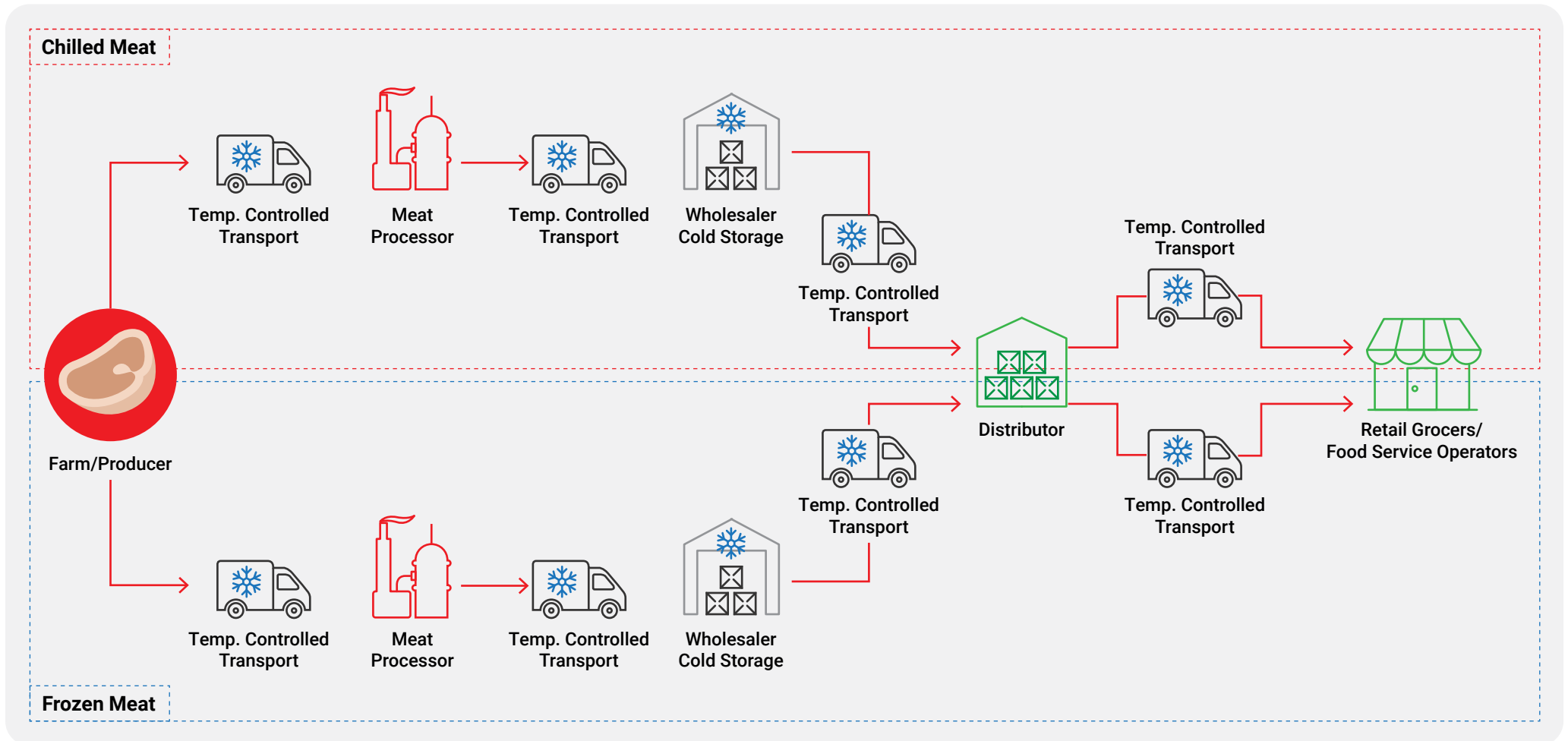
Problem Statement

The Food Safety Modernization Act Surveillance Working Group has a panel of experts who provide guidance to improve foodborne illness surveillance systems in the United States. According to the FSMA Surveillance Working Group 2018 Report that is submitted to the HHS Secretary, 15 pathogens in food items lead to cost of USD 15.5 billion in the United States per year. This includes medical costs, productivity loss, and premature death. [Reference 3]. According to data available from FDA, about 1 in 6 of the US population gets sick by food borne diseases yearly. This is estimated to be fifty million people in the US population.

Similarly, if we consider global figures on a yearly basis, according to WHO estimates, nearly 600 million people fall ill due to food-borne diseases, and 420,000 people die due to food-borne diseases. [reference 2]



Typical Refrigerated Food Supply Chain



In the above diagram, a representative diagram for the cold supply chain is demonstrated using chilled and frozen meat. Chilled meat needs to be stored and transported at a temperature of zero degree to four degrees Celsius in refrigerated containers to avoid the growth of pathogenic bacteria. Similarly, for frozen meat, the ideal temperature conditions are zero degree Fahrenheit (-18 degrees Celsius) or lower.

The required temperature levels can be maintained using vacuum cooling, forced air cooling, or hydro cooling, and a mechanism suitable for the food item will be chosen.

In brief, if we consider the supply chain, the cattle farmers supply the meat processors. The meat processor packages the meat in cases. These multiple cases are enclosed within a pallet as a part of the palletization process. These pallets are loaded on the refrigerated trailers during the loading process. As the trailer moves through the processor, wholesalers, and distributors, the process of checking out and checking in is taken care of. Eventually this is sold out to the retailer and is taken care as Order receival process. The retailers sell to the end consumers.



Challenges in the Refrigerated Food Supply Chain

Businesses involved in the refrigerated food supply chain, are always challenged to mitigate the risk of pathogenic food borne disease and in case there is a suspect or potential pandemic break out, optimize the cost of recalls associated with food items. To achieve this, the businesses must overcome multiple technical challenges, and the following are the most important technical challenges:

- **Multiple entities and disparate systems:** Multiple entities typically get involved in the transport of refrigerated food items from the processor to the retailer. Many of the entities will have their own IT systems to track incoming trailer loads, gate entries, inventory picking against orders, and gate exit shipments. This makes it difficult to get traceability and a sole source of truth.
- **Excursion Management:** Refrigerated food items need to be monitored for temperature throughout the supply chain, from the initial packing to the retailer. If the temperature excursion happens, mechanisms shall be available to immediately alert the entity that is the current custodian of the food pallets and take corrective action.
- **Traceability to an entity:** Traceability refers to the ability to identify either upstream or downstream in the supply chain, the source of the food items, its ingredients. In case there is a disease breakout, or a suspected breakout, it is important to identify the source of the food items that went into the batch or load.
- **Tracking of food cases and pallets:** This is the ability to track the current location and custodian of the food items belonging to a batch.
- **Compliance with regulatory norms:** Food items, while they are processed, transported, or distributed, need to comply with the regulatory norms applicable to the country. E.g., in the United States, the FDA plans to mandate the Food Safety Modernization Act (FSMA).



FSMA 204

FDA's Food Safety Modernization Act (FSMA) is targeted towards preventing foodborne diseases. FSMA has enacted seven major rules and the Food Traceability final rule for Food Traceability is one among them.

The FDA has created a final rule for the traceability of certain foods. The final rule is part of Section 204(d) of the Food Safety Modernization Act (FSMA). Entities who manufacture, process, pack, or hold foods on the Food Traceability List (FTL) must maintain records containing Key Data Elements associated with specific Critical Tracking Events (CTEs) and provide information to the FDA in 24 hours or some reasonable time. The FDA could ask the firms to provide such information in the event of any disease breakout.

The Critical Tracking Events in the final rule are the following:

- Harvesting
- Cooling
- Initial Packing
- First Land Based Receiver
- Shipping
- Receiving
- Transformation

Food items on the traceability list are identified using a Traceability Lot Code (TLC). TLC needs to be assigned to a traceability lot that is within the records of a firm that is doing initial packing, cooling, shipping, receiving, or transformation.

Traceability Lot code needs to be assigned in the following cases:

- Initially pack a raw agricultural commodity other than food obtained from a fishing vessel.
- Perform the first land-based receiving of a food obtained from a fishing vessel; or transform a food.

After a food item is assigned a TLC, all Key Data Elements (KDE) shall be captured along with the TLC at each Critical Tracking Event (CTE) for the food traceability lot. This is used to trace and track food items within the supply chain and provide relevant information to the next party within the food supply chain. Please refer to [Reference 1] for further details.

The FDA has set aside January 2026 for all the concerned entities to comply with the final rule.



Understanding of GS1 Standard

GS1 is a global standards body that develops and maintains supply chain standards.

The GS1 standard is very widely used by the food supply chain business, which provides them a foundation for identifying their products uniquely, capturing them, and sharing the necessary information.

The standard provides a comprehensive set as below:

- **Identify:** GS1 standard for identification
This includes the location, which could be identified by a Global Location Number (**GLN**). The products in the business line have a Global Trade Item Number (**GTIN**) and a Serialized Global Trade Item Number (**SGTIN**). In the logistics trade, container identification is done by the Serial Shipping Container Code (**SSCC**). They can also be identified as a “Global Shipment Identification Number” (**GSIN**). There are many other application identifiers that are part of the standard.
- **Capture:** GS1 standard for barcodes and EPC/RFID
The products could be traced with printed labels on the containers, items, and trailers. These labels are GS1 Barcodes and GS1 EPC/RFID. These are interfaced with hardware components such as Printers for printing labels and reading the labels are done by using handheld readers or fixed readers mounted on the walls. The software component carries the data to the application.

- **Data Exchange:** GS1 Standard for Data Exchange

The information sharing includes the gathered data, which is refined and enhanced with business context to transform it into data that can be shared using standard exchange protocols. This is the transactional data exchanged between two entities in the supply chain.

Master data: This data includes location and product details.

Event data: This data includes the status of the item, product, or container, which is received from the RFID tag read.

- The GS1 Global Traceability Standard is used across the lifecycle of the food supply chain and the events that are generated. The three main levels of identification are
 - » **Class-level identification:** At this level, the object is identified by its product ID
 - » **Batch- or lot-level identification:** At this level, the product has a batch or lot number associated with it
 - » **Instance-level identification:** At this level, the object has a serialized ID



Opportunities for Digitalization

Traceability in the food supply chain is associated with data at each step as the product moves across different tenants like the manufacturer, wholesaler, distributor, and retailer. As required in FSMA 204 requirements, Critical Tracking Events and Key Data Elements can be captured using the GS1 standard.

- Critical Tracking Events (CTEs) - These are the events that occur to traceable objects during their lifecycle, such as creating, palletizing, trailer loading, exit, entry, receiving, and disposing.
- Key Data Elements (KDEs) - These are the pieces of data that describe the actual instances of CTEs. The data can be the combination of the class and batch level identification with the location and process involved with it.

For more information, please check [Reference 4].

To comply with the FSMA traceability rule and improve visibility and traceability in the supply chain, “farm to fork” digitalization will need to be adopted by different firms in the supply chain.

The digitalization solution will involve the following:

- Digital identities
 - » Digital identities will need to be assigned for a specific food item or SKU in the lot and for the entire lot. Digital identity can be provided by different means, such as barcodes, QR codes, and RFID tags.
- Sensors that can be associated with food cases allow for real-time monitoring of temperature, humidity, and other ambient parameters that can impact food quality.
- Cloud hosted service layer to capture Critical Tracking Events as food items pass through different entities in the supply chain, from the food processor to the retailer.
- Mobile apps, along with readers for different user personas in the supply chain (such as drivers and warehouse workers), capture tracking events and provide excursion alerts for them to take initiative-taking, preventive steps.
- Web portals that can be used by different entities in the supply chain for viewing reports, viewing shipment status, alerts etc.



References:

1. FSMA Final Rule: Requirements for Additional Traceability Records for Certain Foods (<https://www.fda.gov/food/food-safety-modernization-act-fsma/fsma-final-rule-requirements-additional-traceability-records-certain-foods>)
2. <https://www.who.int/activities/estimating-the-burden-of-foodborne-diseases>
3. <https://www.cdc.gov/ddid/docs/BSC-OLD-FSMA-SurvWG-2018-AnnualReport-508.pdf>
4. GS1 Standard (<https://www.gs1.org/fresh-foods>)

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.

About the Authors

Vivek Madathil, Chief Architect in the Digital Services Business Line, has close to 25 years of industry experience and has experience in defining solutions, offerings, and providing thought leadership in cloud, data analytics, and application development to customers from various verticals such as automotive, industrial, and public transportation.

Shweta Naik, Project Manager in the Digital Services Business Line, has close to 16 years of industry experience and has experience managing projects in IoT solutions, automotive, healthcare, and industrial.





Food Cold Chain: Challenges and Opportunities for Digitalization

marketing@sasken.com | www.sasken.com

USA | FINLAND | GERMANY | JAPAN | INDIA

© Sasken Technologies Ltd. All rights reserved.

Products and services mentioned herein are trademarks and service marks of Sasken Technologies Ltd., or the respective companies.